

Conditional Probability

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Introduction

- We've been working with probabilities of events.
- Today we introduce **conditional probability**: the probability of something *given* that something else is true.
- This is one of the most important and widely-used ideas in the course.

Associated Reading

Odds and Ends, Chapter 6

What Conditional Probability Is

- Sometimes we don't just care about how likely something is.
- We care about how likely it is *given* that some other thing has happened or will happen.

This matters for:

- Planning for the future given new information.
- Computing overall probabilities using partial information.
- Updating our beliefs when we learn something new.

The Formula

$$\Pr(A \mid B) = \frac{\Pr(A \wedge B)}{\Pr(B)}$$

- The left-hand side means “the probability of A given B .”
- The right-hand side: the probability of both A and B being true, divided by the probability of B .

Why This Formula?

Intuitively: conditioning on B means we restrict attention to the cases where B is true.

- The probability of A given B is the share of the “ B -worlds” in which A is also true.
- We take the probability of both A and B , and divide by the probability of B to rescale.

One pitfall: the formula requires $\Pr(B) > 0$.

If $\Pr(B) = 0$, then $\Pr(A \mid B)$ is simply **undefined** (not zero, not a special number).

There's just no such thing as 'the probability of A given B ' when B has zero probability.

Conditional Probability in Trees

Tree diagrams are a natural home for conditional probabilities.

In a probability tree, branches represent possible outcomes. The number on each branch is the probability of that outcome **conditional on all previous branches**.

The first branch is the only one where the probability is *unconditional*. There are no prior branches to be conditional on.

To find the probability of a complete path, **multiply** the branch probabilities along it.

Updating

There is a key philosophical claim we will use throughout:

$$\Pr_B(A) = \Pr(A \mid B)$$

The *new* unconditional probability of A after learning B equals the *old* conditional probability of A given B .

This is not a mathematical theorem. It is a claim about how rational agents *should* respond to learning new information. It is very widely accepted.

- It tells us that conditional probability isn't just a mathematical curiosity.
- It tells us how to *use* probability to represent learning and reasoning under uncertainty.
- Whenever we learn something, we use the conditional probability formula to update.

Order Matters

In general, $\Pr(A \mid B)$ and $\Pr(B \mid A)$ are **different quantities**.

Example from the book: most university students are young, but most young people are not university students.

- $\Pr(\text{young} \mid \text{university})$ is high.
- $\Pr(\text{university} \mid \text{young})$ is low.

Switching the order of conditioning changes the meaning and the number.

A Common Confusion

The confusion between $\Pr(A \mid B)$ and $\Pr(B \mid A)$ is one of the most frequent mistakes in everyday reasoning about probability.

It has a name: the **transposition fallacy** (or “the prosecutor’s fallacy” in legal contexts).

Example: “The probability that an innocent person would leave this evidence is very low. Therefore the probability this person is innocent is very low.” These are **not** the same probability.

Keeping the order straight is essential. On Wednesday we’ll go into this in more detail.

Independence

A and B are **independent** if and only if

$$\Pr(A \mid B) = \Pr(A) \quad (\text{and } \Pr(A) > 0)$$

That is: learning B makes no difference to A 's probability.

Equivalently: $\Pr(A \wedge B) = \Pr(A) \times \Pr(B)$.

This is the “independence means multiply” mantra from §6.5.

Three Facts About Independence

The book (§6.5) establishes three useful facts:

1. **Multiply:** A and B are independent iff $\Pr(A \wedge B) = \Pr(A) \Pr(B)$ (assuming both have positive probability).
2. **Symmetry:** if A is independent of B , then B is independent of A .
3. **Negations:** if A is independent of B , it is also independent of $\neg B$ (assuming $\Pr(\neg B) > 0$).

- B might be a possible **cause** of A .
- B might be a possible **preventer** of A .
- B might be a common effect of a frequent **cause** of A .
- B might be a common effect of a frequent **preventer** of A .

- B being true could tell you that a source which also predicts A is more reliable than you thought.

For example: two eyewitnesses both reporting the same crime gives more evidence than one, but only if the reports are independent. If they talked to each other first, they are not independent, and we should discount accordingly.

Two Important Real-World Facts

1. In reality, strict independence almost never obtains.
2. In practice, it's very often useful to *assume* independence for modelling purposes.

These are consistent, but be careful. Sometimes assuming independence is like assuming relativistic effects don't matter for bridge design.

Sometimes it's fine. The context determines which.

An Interesting Example

Two fair dice, one red and one blue, are tossed. Let:

- A = the sum of the two dice is 7
- B = the red die shows a 2

Are A and B independent?

Surprisingly: **yes**.

Checking Independence

If the red die shows 2, the blue die must show 5 for the sum to be 7. That's 1 out of 6.

$$\text{So } \Pr(A \mid B) = \frac{1}{6}.$$

And the unconditional probability of getting a sum of 7 is: the pairs (1,6), (2,5), (3,4), (4,3), (5,2), (6,1). That's 6 out of 36 = $\frac{1}{6}$.

So $\Pr(A \mid B) = \Pr(A) = \frac{1}{6}$: they are independent.

Worked Exercises

Exercise 4: Lead Exposure in Elizabeth

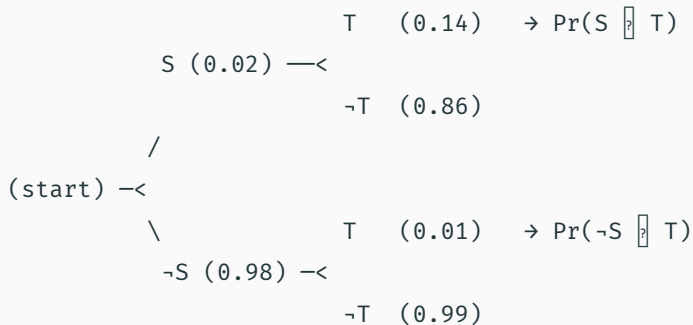
In the city of Elizabeth, 2% of children live in Southside (near chemical plants). Of those, 14% have toxic levels of lead exposure. Elsewhere in the city, only 1% of children have toxic exposure.

Let S = child lives in Southside, T = child has toxic lead exposure.

We are told:

$$\Pr(S) = 0.02, \quad \Pr(T \mid S) = 0.14, \quad \Pr(T \mid \neg S) = 0.01$$

Exercise 4: Drawing the Tree



Exercise 4a: $\Pr(S \mid T)$

Question: What is the probability a randomly chosen child lives in Southside *and* has toxic exposure?

This is the top complete path in the tree: multiply along the branches.

$$\Pr(S \wedge T) = \Pr(S) \times \Pr(T \mid S) = 0.02 \times 0.14 = 0.0028$$

About 0.28% of all children (a small number), but the concern is whether it is *concentrated*.

Exercise 4b: $\Pr(T)$

Question: What is the probability a randomly chosen child has toxic exposure?

There are two ways to have toxic exposure: via Southside or via elsewhere.
Sum the two paths:

$$\Pr(T) = \Pr(S \wedge T) + \Pr(\neg S \wedge T)$$

$$= \Pr(S) \Pr(T \mid S) + \Pr(\neg S) \Pr(T \mid \neg S)$$

$$= (0.02)(0.14) + (0.98)(0.01) = 0.0028 + 0.0098 = 0.0126$$

Exercise 4c: $\Pr(S \mid T)$

Question: Given a child has toxic exposure, what is the probability they live in Southside?

Apply the definition of conditional probability:

$$\Pr(S \mid T) = \frac{\Pr(S \wedge T)}{\Pr(T)} = \frac{0.0028}{0.0126} = \frac{28}{126} = \frac{2}{9} \approx 0.222$$

Even though only 2% of children live in Southside, roughly **22%** of children with toxic exposure live there. The risk is heavily concentrated.

Notice: we needed $\Pr(T \mid S)$ to find $\Pr(S \mid T)$. Order matters. This is exactly the inversion problem.

Exercise 13: Apples and Buckets

4 apples are each placed at random, independently, into one of 10 buckets. Let B_{ij} be the proposition that apples i and j end up in the same bucket.

What is $\Pr(B_{12})$?

Apple 1 goes somewhere. Apple 2 must land in that same 1 bucket out of 10.

$$\Pr(B_{12}) = \frac{1}{10}$$

By symmetry, $\Pr(B_{ij}) = \frac{1}{10}$ for any pair i, j .

Exercise 13a: Is B_{12} independent of B_{34} ?

B_{12} : apples 1 and 2 in the same bucket. B_{34} : apples 3 and 4 in the same bucket.

These involve completely *different* apples. Knowing where 1 and 2 landed tells us nothing about where 3 and 4 went.

Formally: $\Pr(B_{12} \wedge B_{34}) = \frac{1}{10} \times \frac{1}{10} = \frac{1}{100}$

And $\Pr(B_{12}) \times \Pr(B_{34}) = \frac{1}{10} \times \frac{1}{10} = \frac{1}{100}$. $\square$

Yes, B_{12} and B_{34} are independent.

Exercise 13b: Is B_{12} independent of B_{23} ?

B_{12} : apples 1 and 2 together.

B_{23} : apples 2 and 3 together.

These share apple 2, so there is a logical relationship: if both hold, then 1, 2, and 3 are *all* in the same bucket.

Does knowing B_{12} change the probability of B_{23} ?

Apple 3 is placed *independently*. Knowing which bucket apples 1 and 2 share doesn't change where apple 3 goes. It's still equally likely to land in any of the 10 buckets.

So $\Pr(B_{23} \mid B_{12}) = \frac{1}{10} = \Pr(B_{23})$. **Yes, they are independent.**
(Perhaps surprising)

Exercise 13c: Every Pair?

The same reasoning extends to all pairs of B_{ij} propositions:

- For **non-overlapping** pairs (like B_{12} and B_{34}): different apples, clearly independent.
- For **overlapping** pairs (like B_{12} and B_{23}): the shared apple's placement is already fixed in both events, so the remaining apple still lands uniformly at random.

Yes, every pair of B_{ij} propositions is independent.

Exercise 13d: Every Trio?

Now consider the trio $\{B_{12}, B_{23}, B_{13}\}$. We've just established that every *pair* in this trio is independent.

For three-way independence we also need:

$$\Pr(B_{12} \wedge B_{23} \wedge B_{13}) = \Pr(B_{12}) \Pr(B_{23}) \Pr(B_{13}) = \frac{1}{1000}$$

But notice: if B_{12} and B_{23} both hold, then apples 1, 2, 3 are all in the same bucket. This **guarantees** B_{13} .

$$\Pr(B_{12} \wedge B_{23} \wedge B_{13}) = \Pr(B_{12} \wedge B_{23}) = \frac{1}{100}$$

But $\frac{1}{100} \neq \frac{1}{1000}$. This trio is NOT three-way independent.

Pairwise independence does **not** guarantee three-way independence.

This is exactly the point the book makes in §6.5 with the seasons example: a collection can be pairwise independent while the whole collection fails the joint multiplication rule.

It also shows something conceptually important: B_{12} , B_{23} , B_{13} are logically connected (any two imply the third), but that logical connection is *not* detectable from any single pair of them probabilistically.

- We will look at how to **invert** conditional probabilities.
- Given $\Pr(A \mid B)$, how do we work out $\Pr(B \mid A)$?
- This turns out to be one of the most important skills in applied probability.